

Durgansh Dixit

Machine Learning/Artificial intelligence Engineer • saniket919@gmail.com • +91-9997871742

Education

Banaras Hindu University

Masters in Computer Applications

Relevant Coursework: Machine learning , Deep learning, Natural language processing.

Varanasi , India

8.7 CGPA , 2021-23

Ch. Charan Singh University

Bachelors in Computer Applications .

Meerut , India

2016-2019

Experience

Aavaaz Inc.

VP of AI

Chicago Illinois, United States.

April 2024 - Present

- Developed end-to-end automatic speech recognition (ASR) systems using transformer architectures, achieving a 45% reduction in Word Error Rate (WER) on multilingual datasets.
- Implemented and optimized neural text-to-speech (TTS) models using Tacotron2 and FastSpeech2, reducing inference latency by 30% while maintaining voice quality.
- Led research in neural machine translation, developing attention-based architectures that improved BLEU scores by 2.5 points across low-resource language pairs.
- Fine-tuned large language models (BERT and Llama) for cross-lingual tasks, implementing novel pre-training strategies that improved downstream performance by 18%.

AiVerbalyze Technologies Pvt. Ltd.

Machine Learning Engineer

Mumbai , India.

July 2023- March 2024

- Optimized and Quantised TTS and VC frameworks like NaturalSpeech, FastSpeech, VITS through Optimum, caching mechanisms and asynchronous functions, achieving a significant latency of only 1.5 seconds.
- Engineered and deployed robust TTS and VC models, leveraging tools like Git and Django, and deploying on Azure cloud infrastructure, GCP.
- Led Natural Language Processing team in analyzing conversational data, utilizing advanced technologies for comprehensive understanding.
- Facilitated in the development of a new novel modified compression techniques tailored for neural audio models increasing efficiency of search by a factor of 40 %

Technical Skills

Languages: Python, Java, C++, Javascript, Dart.

Libraries: transformers, pytorch, tensorflow, huggingface, wandb, langchain etc.

Tools: Azure, GCP, jupyter, conda, Git, Docker, Kubernetes etc.

Techniques: CoT, ReAct, Reinforcement Learning, Generative Adversarial Networks, Natural Language Processing, Computer Vision, Robotics.

Recent Projects

LawyerGPT

- Fine tuned llama2-7b using LoRA on indian legal dataset.
- Migrated the System from RAG (RetrievalAugmentedGeneration) to fine-tuned system.
- Used Sentencetransformers, Hugginface, Pytorch, transformers to develop the project.
- Secured commercial adoption of the model with a top legal firm in India.